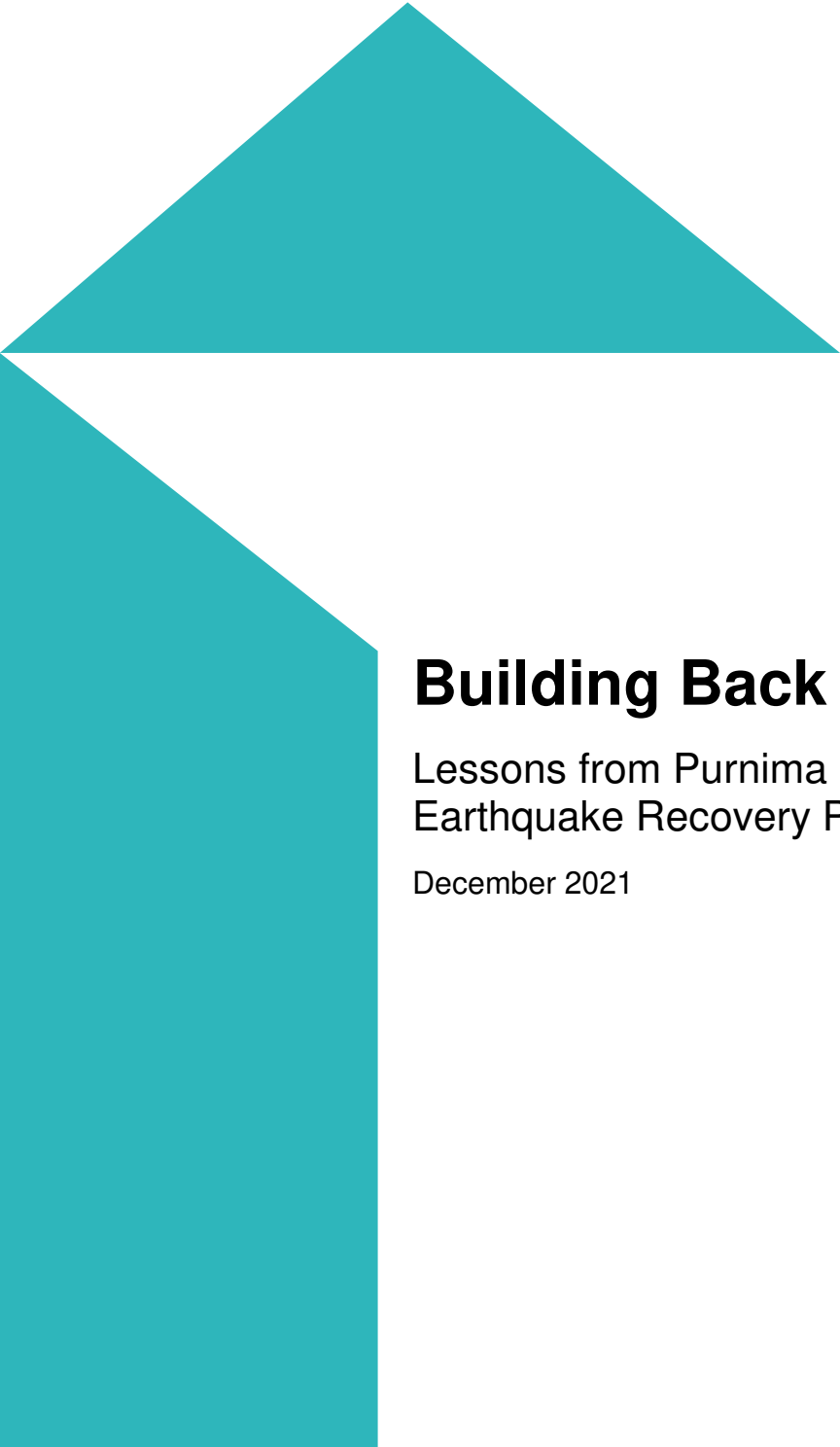




Building Back Better

Lessons from Purnima - a UK Aid-Funded Post-Earthquake Recovery Programme in Nepal

December 2021

Mott MacDonald
22 Station Road
Cambridge CB1 2JD
United Kingdom

T +44 (0)1223 463500
mottmac.com

Building Back Better

Lessons from Purnima - a UK Aid-Funded Post-Earthquake Recovery Programme in Nepal

December 2021

Issue and Revision Record

Revision	Date	Originator	Checker	Approver	Description
V3	25.11.2021	S Watson	K Wagle	S Howarth	Purnima Lessons

Document reference: 389785 |

Information class: Standard

This document is issued for the party which commissioned it and for specific purposes connected with the above-captioned project only. It should not be relied upon by any other party or used for any other purpose.

We accept no responsibility for the consequences of this document being relied upon by any other party, or being used for any other purpose, or containing any error or omission which is due to an error or omission in data supplied to us by other parties.

This document contains confidential information and proprietary intellectual property. It should not be shown to other parties without consent from us and from the party which commissioned it.

Contents

1	Introduction	1
2	The Project	2
3	Resilience of Institutions	3
3.1	Political economy analysis	3
3.2	An integrated approach	4
3.3	Capacity development	5
3.4	Transparency, accountability and inclusion	5
4	Resilience of People	6
4.1	Understanding vulnerability	6
4.2	A system wide approach	7
4.3	Building local ownership and awareness of vulnerability	7
4.4	Strengthening service delivery for vulnerable groups	7
4.5	Housing reconstruction for vulnerable groups	8
5	Resilience of Infrastructure	9
5.1	Infrastructure quality and delivery	9
5.2	Sustainability and maintenance	10
5.3	Gender equality and social inclusion	10
5.4	Technical delivery	Error! Bookmark not defined.

1 Introduction

This paper identifies and discusses the lessons from Purnima - a UK Aid funded post-earthquake recovery programme in Nepal. Drawing on learning from the programme's three intervention areas – resilience of institutions, people, and infrastructure – it seeks to provide lessons to government and donors for future post-disaster recovery programmes in Nepal and elsewhere.

Most of the world's population now lives in areas affected by natural hazards, and this proportion is expected to rise. Globally, one in three people is exposed to earthquakes, a number which has almost doubled in the past 40 years¹. In this context, the lessons of 'building back better' after the 2015 Nepal earthquake are relevant to agencies and governments seeking to respond to natural disasters and strengthen the resilience of at-risk communities, institutions and infrastructure.

The 7.8 and 7.3 magnitude earthquakes that struck Nepal caused more than 9000 deaths and destroyed over 600,000 homes. Incomes plummeted as people struggled with injury, the loss of loved ones, and destruction of their houses and livelihoods. Relief and reconstruction efforts were hampered both by the remote, mountainous terrain and the ingrained socio-economic fragility of the affected populations.

Nepal's recovery was further complicated by a huge political shake-up: after years of wrangling, the country adopted a new constitution just three months after the disaster, which included a major devolution of political and administrative powers to the local level. Although this decentralisation of authority brought a sense of optimism to earthquake-affected areas, the new governments were inexperienced, severely understaffed and had few resources.

Simultaneously, there was a huge surge in international aid, donor activities and NGOs seeking to support the recovery and humanitarian response. While critical in saving lives and alleviating suffering, this threatened to weaken local capacity and agency, and undermine the institution building needed for more sustainable development. Five years after the earthquake, as recovery activities were beginning to look ahead and focus on longer term development, the global Covid-19 pandemic plunged Nepal into another severe economic shock.

Here we present an overview of Purnima, a UK Aid-funded post-earthquake recovery programme, how it sought to respond to these challenges in four districts in Nepal, and the lessons learnt about how to build back better institutions, livelihoods and infrastructure in a fragile post-earthquake environment.

¹ [Atlas of the Human Planet 2017 – how exposed are we to natural hazards?](https://ec.europa.eu/science-hub/en/atlas-of-the-human-planet-2017-how-exposed-are-we-to-natural-hazards/) | EU Science Hub ([europa.eu](https://ec.europa.eu/science-hub/en/))

2 The Project

Purnima is a five-year programme (2018-2022) funded by UK Aid and managed by Mott MacDonald. The objective is long-term positive change in the lives of people in earthquake-affected regions of Nepal.

The programme works in four of the worst affected districts (Gorkha, Dhading, Nuwakot, and Rasuwa) to build:

- Resilience of institutions via capacity development of local government
- Resilience of people via socio-economic support for vulnerable groups
- Resilience of infrastructure via direct physical reconstruction of priority rural infrastructure.

Our vision for the programme was not just to reconstruct earthquake-damaged infrastructure to help the affected population. The situation in post-earthquake Nepal provided an opportunity to 'build back better', not only infrastructure but also social inclusivity and economic opportunities. Our objective was to ensure that no one was left behind in the reconstruction process, and we targeted the most vulnerable population in all activities. The infrastructure was selected through a participatory planning process, by working closely with Palikas and wards.

The project aimed to strengthen the ability of local government and communities to deliver inclusive services and infrastructure at a local level. We linked our interventions in a systematic and coherent way to rebuild rural infrastructure and increase the socio-economic wellbeing and resilience of the most vulnerable groups. The specific vulnerable groups that we targeted under the 'leave no one behind' component included persons with disabilities, the elderly, vulnerable single women (especially Dalit), Internally Displaced Persons (IDPs) after resettlement, and extremely poor and food insecure people from remote areas.

We sought to ensure that our activities were consistent with local needs and priorities, and coordinated with other government and partner programmes, so that we worked together towards a common vision. We worked with government especially with local governments, civil society, NGOs, private sector and development partners.

3 Resilience of Institutions

Institutional resilience is the capacity to effectively and efficiently deliver and improve over time while ensuring **trust, legitimacy and credibility**; as well as the ability to **manage shocks and change**. Beyond mere delivery capacity, an institutions resilience is therefore dependent on its accountability, its level of inclusivity or exclusivity, and its norms and networks². According to the OECD, “for development co-operation actors, promoting institutional resilience means **staying the course over the long term** and prioritising the use of **local knowledge, experiences and resources**. It also implies a decisive shift... towards locally embedded, iterative interventions that promote institutional responsiveness and flexibility³.”

The importance of institutional resilience was demonstrated in Nepal only three months after the earthquake, when the government adopted a federal system and devolved a raft of powers to the local and provincial levels. Political uncertainty and an international economic blockade disrupted the response to the earthquake, but the first local elections in 20 years were held in 2017. Two years later, the Covid-19 pandemic struck. Nepal’s local governments, whilst still weak and nascent, has had to cope with numerous shocks and changes and continue to deliver and improve. This presented enormous challenges but also entry points for change.

Elected in 2018, the new local-level governments (*Palikas*) were enthusiastic but poorly resourced, understaffed and under skilled. Their responsibilities now included management of local services, development plans and disaster management. Purnima worked with 22, mostly rural, Palikas in the districts worst affected by the earthquake to support inclusive governance and planning, and to strengthen institutional resilience for better service delivery. This section describes the lessons learnt.

3.1 Political economy analysis

Purnima’s three political economy analyses covering macro and micro levels as well as details of the planning process provided crucial information for planning the interventions for institutional resilience.

- **Understand the barriers to inclusive recovery:** The infrastructure sector and its intersection with government is dominated by informal ‘rules of the game’: it operates on personal relationships, patronage systems - be they party political, kin, or other relational dimensions – and frequent collusion between actors, coupled with distortion of fair practices in the award of contracts. Many politicians have businesses interests in the sector. Those currently benefiting have high incentives to adjust and re-establish systems to their benefit.
- **A unique opportunity in time:** It is very hard to shift such normative and all-pervasive behaviours in any environment, however a combination of high expectations of improved development from the public, increased political will to deliver by elected local officials, and the broad disruption of current systems provides a chance for influencing norms, behaviours and processes.
- **Strengthen social participation and accountability:** entrenched social inequalities expressed through caste, ethnicity and gender are a persistent barrier to inclusive development. Improving civil society participation and accountability mechanisms is vital. At government level, this can be achieved via dissemination of information, consultation and

² [Supporting institutional resilience | Development Co-operation Report 2020 : Learning from Crises. Building Resilience | OECD iLibrary \(oecd-ilibrary.org\)](#)

³ [Supporting institutional resilience | Development Co-operation Report 2020 : Learning from Crises. Building Resilience | OECD iLibrary \(oecd-ilibrary.org\)](#)

participation, independent complaints/feedback mechanisms, and implementation of local governance self-assessments.

- **Strengthen local autonomy and revenue sources.** Palikas have few sources of internal revenue and are largely reliant on federal grants. The main local source for some palikas is from quarrying riverbed materials which is environmentally risky. Purnima supported Palikas to develop a Revenue Improvement Action Plan (RIAP) to identify, raise and administer a broader base of local taxes more responsibly. This has allowed Palikas to finance local development initiatives and increases their accountability to local communities.

3.2 An integrated approach

Institutional resilience is not an isolated characteristic but the product of many aspects of government capacity: focusing on one aspect of delivery is unlikely to achieve results; rather, a holistic approach is required. Purnima promoted institutional resilience through an integrated approach combining capacity development of local government, delivery of infrastructure, socio-economic support and private sector development.

- **Donors should combine capacity development with hardware:** The social capital we derived from delivering concrete results in infrastructure delivery, which is a key priority for local communities and their representatives, gave us credibility with palikas and the influence to push for more politically sensitive institutional development such as greater transparency and inclusion. This integrated approach also created opportunities to build local capacity by transferring learning and ownership. For example, we performed in depth vulnerability assessments to identify people who needed livelihoods support in each Palika. This data was formally handed over to the governments who are now using and updating it for their own planning and budgeting.
- **Progressively transfer ownership:** To meet the post-earthquake needs, Purnima initially delivered reconstruction and socio-economic support through NGOs, who had the capacity and experience and were able to mobilise quickly. They also provided a counterweight to government in terms of inclusion of vulnerable groups, independence and transparency. We then worked to involve Palikas in the inclusive selection, co-financing, delivery and monitoring of infrastructure and other services financed by the project to strengthen their capacity, agency and credibility with local populations. This included capacity development on climate and disaster resilience and adoption of more transparent online public procurement mechanisms. Then, we focused on the Palikas' ability to deliver infrastructure and services with their own resources.
- **Social inclusion is central to resilient delivery:** we supported Palikas to produce costed annual development plans through community consultation. However, Palikas did not consider inclusion a priority. So, we helped each Palika to carry out an audit of gender equality and social inclusion (GESI) and to formulate a GESI policy. This gave a stronger voice to women and disadvantaged groups during consultations, strengthened representatives' understanding of their roles and responsibilities, and created awareness that inclusion is central to good local planning. As a result, Palika budgets for activities related to vulnerable groups increased.⁴
- **Creating links between different local actors builds resilience.** we facilitated discussions between local market actors, small businesses and local government which led to greater understanding and streamlining of regulations and red tape and helped develop agricultural supply chains and financial services for vulnerable groups.

⁴ For example, in Siranchowk the budget increased to 10% (NPR 71.3 million) from 2.93% (NPR 10.6 million)

3.3 Capacity development

Capacity development was a central component of Purnima's work on promoting institutional resilience. The way we delivered the services was important for the success and sustainability of this activity:

- **Embedded support and teamwork build trust:** Purnima supported local governments through embedded capacity development, meaning a Governance Expert and an Engineer were seconded to each Palika office. This permanent presence was key to gaining trust and confidence. Given the remoteness of many of the offices, it also demonstrated our commitment to the local communities and helped ensure we understood local realities and institutional dynamics. Building a sense of teamwork with Palika staff was also key: since Palikas were understaffed and lacked important skills, initially our staff helped them perform core functions. Over time, as Palika capacity improved, capacity development became the dominant activity.
- **Careful sequencing:** It was essential that capacity development activities and training were carefully sequenced to match the phases of the Palikas own planning and budgeting cycles.
- **Results-based contracting of training can improve quality:** Providing good quality, timely and effective training on the skills that local governments need in very remote areas required logistical flexibility, strong technical knowledge and excellent facilitation skills. We found that private sector providers, contracted via competitive tender and using results-based contracting, were the most effective delivery partner for technical trainings.
- **Time and commitment:** Building relationships, identifying what worked, and scaling up took time and commitment. Three to five years of working with the governments was the minimum time needed to make sustainable gains.

3.4 Transparency, accountability and inclusion

Ensuring transparency, accountability and inclusion is crucial for institutional resilience. Purnima addressed this through:

- **Focus on the lowest layers of government:** Wards are the closest administrative unit to the Palika residents and are responsible for understanding and addressing citizens' needs for effective service delivery. Purnima initially concentrated on strengthening systems and structures in the Palikas, which are the top level of local government. Once good progress had been achieved at the apex level of local government, Purnima moved its focus of attention to the logistically more complex task of building capacity of the large number of Wards, as they are better placed to communicate to and represent local communities, including the most marginalised. Unfortunately, however, the intervention of the COVID-19 pandemic limited the amount of work that could be done at this level.
- **Promote community voice and empowerment:** our mandate was to focus on capacity development of the local government, but to build genuine institutional resilience, people and communities must be empowered to know their rights and hold their representatives to account. We improved transparency and accountability on the 'supply' side (for example, through increased use of Palika websites to inform on projects, public audits, e-bidding and community consultations), but future programmes should also address the 'demand side' so that communities and minorities can understand and articulate their rights.
- **Special trainings to marginalised representatives:** Various studies conducted by Purnima (GESI Audit, LISA study, Perception Survey of GESI in palikas) pointed out that women and marginalised representatives lag behind in knowledge, awareness, and empowerment. This is a large demographic group that can have strong influence in steering development in palikas. Though the constitution has given them decision making authority, it is barely exercised. They require special, focused trainings that are designed to meet their needs and is paced according to their requirements. Without their full participation, development will be disproportionate.

4 Resilience of People

When assessing the destruction caused by a natural disaster, the focus is often on buildings, infrastructure, and lost output. This approach can overlook the relative losses experienced by poor and marginalised people, whose consumption is closer to subsistence level and whose livelihoods are dependent on much smaller assets. The loss of a goat or a crop cycle might not make headlines but can have a devastating impact on the livelihood and wellbeing of someone who is very poor. It has been shown that poor people suffer more from natural disasters: they cannot rely on savings to smooth the impacts, they lose a greater share of their income, their dwellings are more fragile, and they may need more time to recover and reconstruct.⁵

‘Resilience of people’ means therefore ensuring that assets and livelihoods become less vulnerable to future shocks, promoting livelihoods that are sustainable, and ensuring that nobody – including the most vulnerable - is left behind in the recovery process. Following the earthquake in Nepal, household debt burdens and other negative coping strategies increased, with associated reductions in well-being, increased social exclusion and disempowerment, and continued food insecurity.

Purnima provided socio-economic support to strengthen the resilience of five groups of vulnerable people – those with disabilities, the elderly, single women, internally displaced persons and the extremely poor. We also supported over 600 small businesses to develop the local economy and market supply chains. This section discusses the lessons learnt.

4.1 Understanding vulnerability

Dedicate time and resources to understand and identify who is vulnerable: When we began there was a scarcity of data on who was vulnerable and where, why people were vulnerable, and how to improve their livelihoods and resilience. It was essential to establish clear, in-depth and localised data on vulnerable people to understand what interventions and support would reach them best. An initial assessment identified five groups as being most vulnerable – those with disabilities, the elderly, single women, internally displaced persons and the extremely poor. But finding those individuals was a huge challenge: many were illiterate, living in remote locations, had limited mobility and lacked the knowledge and confidence to seek help. Through extensive focus group discussions with communities at the level of individual settlements, we identified and mapped vulnerable households in four districts. We then conducted a survey of each household using electronic survey tools to better understand their incomes, skills, socio-economic status and housing status. Developing this detailed understanding of vulnerable households was crucial to be able to provide the right support to the right people.

Balance targeted support with scalability: Once we knew whether people had land available for agricultural activities, their level of mobility, their level of access to markets, their skills, and level of family support, we supported them with livelihood skills and assets adapted to individual requirements. However, in some cases this led to an overdiversification of livelihood sectors, which limited scalability and our ability to work on service and market linkages. We therefore changed approach and focused on a smaller number of sectors which had greater scalability, but which we knew had low entry barriers for all groups, including vegetable farming, poultry farming, and goat keeping. Where needed, we adapted these sectors to individuals – for example, by making a poultry farm wheelchair accessible.

⁵ [211003ovEN.pdf \(worldbank.org\)](#)

4.2 A system wide approach

Vulnerable people face multiple barriers which limit their social and economic development. They lack capacity, have limited productive resources, and also have limited access to services. Likewise, there are various social norms and restrictive policies, which limit their social and economic development. Hence, a system wide support; support at different levels, is required to remove the barriers and generate effective and sustained results for them. In practice, this meant working at three levels:

1. at household level to improve livelihoods by providing training and assets for income generating activities.
2. building ownership and awareness in local government to respond to vulnerable groups.
3. Improving market linkages and access to services for vulnerable groups.

4.3 Building local ownership and awareness of vulnerability

Strengthen institutional data and understanding: once we had completed our vulnerability assessment, we formally handed the data sets over to the Palikas, who now publish it on their websites, update it themselves, and use it to distribute identification cards (vital for accessing services), provide social security allowances, and target socio-economic support to vulnerable people. As an example of strengthening institutional resilience, they used the same data to provide services and support to vulnerable households during the pandemic.

Alignment towards government policies and priorities: we aligned our support to local government policies and commitment towards vulnerable people. Palikas have responsibility for specific vulnerable groups (disabled people, senior citizens) per the constitution and as outlined in the local government operation act, and a responsibility to ensure inclusion in planning and delivery of local services. We helped Palikas draft and adopt policies for gender equality and social inclusion, guidelines for disability inclusion, and livelihood recovery plans for internally displaced persons. We also built awareness in Palikas on their commitments towards vulnerable people by facilitating Local Governments Institutional Self-Assessment (LISA), which allowed Palikas to assess their own capacity gaps on these responsibilities. As a result, Palikas significantly increased their budgets for socio-economic support to vulnerable groups (to NRs173m in 2021/22).

4.4 Strengthening service delivery for vulnerable groups

Linkage with service providers: During the first two years of the project, we provided training and investment to individual households to improve their livelihoods. However, the wider market barriers for vulnerable households meant that the return on investment was relatively small. Households had low bargaining power, lacked information on how to sell to markets and limited access to private services to support their activities. Therefore, we mapped existing service providers, strengthened them where needed and linked them to vulnerable households and organised groups of vulnerable smallholders. For example, goat raising is among the most profitable activities for vulnerable people. However, the death of a goat can push people back into debt and poverty. Hence, we helped households to insure more than 1,400 goats in collaboration with the Agriculture and Livestock section of Palikas. We also supported the signing of an agreement between the Palika and insurance companies for better insurance coverage.

Building on existing local resilience: Mapping and mobilisation of existing service providers was very effective for providing sustainable services to vulnerable people. There were service providers, in some form, in even the remotest project areas, and it is easier for vulnerable people to deal with local providers. Some required capacity building support before mobilisation. For others, we just needed to facilitate the linkage with vulnerable people. We supported three

groups in Dhading to establish goat resource centres to provide breeding, treatment and marketing services to goat-raising smallholders.

Overcome administrative barriers: Cumbersome policies and procedures limited the access of vulnerable people to services, even depriving them from the benefits of targeted schemes. The business registration process is too complicated for many small businesses and marginalised/vulnerable people who have little education, to navigate. It can take months and cost thousands of rupees besides the designated fees, which is why most people run their business and activities informally. Our business help desk, run by the local chamber of industry, helped more than 1000 people, mostly women and people from marginalised communities, to register their business and activities. This helped them to access loans and government support for small businesses.

The journey to resilience is gradual: For many vulnerable people, the journey to self-resilience is gradual as many of them are not even economically active. Hence, we should start preparing them for economic activities by improving their self-esteem, skills and providing start up support. Then, their economic activities need to be protected and promoted through inclusive policies and affirmative action till they are able to compete with others in the market – which could be done through better organisation, upgrading their skills and products. When they are able to compete in the market and able to negotiate with the other market actors in the win-win term, they should be linked with the existing market actors for services. The vulnerable people need a lot of hand holding during the process – heavy handed support in the earlier stages and lighter support during later stages.

4.5 Housing reconstruction for vulnerable groups

We supported vulnerable households in the entire reconstruction process. Purnima's embedded engineer in the respective Palika areas conducted periodic monitoring and the supervision of the activity to ensure the quality of the construction and expedite the construction activity to complete on the planned timeframe.

Vulnerable people need livelihood support as well as safe housing: Livelihood support should go hand in hand with housing reconstruction support. Such support not only provides them with a secure livelihood but can also contribute to a house reconstruction and prevent them from falling back into a debt trap.

Vulnerable people need more flexibility in house design as the prototype designs pose limitations to households who lack suitable land. The households can go for a customized design but the process is too complicated for vulnerable households to navigate. Likewise, accessibility aspects should also be considered in the housing design, which is lacking in some prototype designs. There is also more scope to promote the use of other local materials and traditional features can make the housing more affordable.

Vulnerable people are information-poor and lack awareness about quality and safety standards and compliance. They need rigorous support to assure quality and compliance from the very beginning. Correction in already built structures is expensive and many households cannot afford this.

5 Resilience of Infrastructure

The provision of infrastructure services to communities during and immediately following disasters is critical. In Nepal, many rural water supply sources shifted or dried up after the earthquake, depriving communities of water access. This was life threatening during the immediate post-earthquake period but also hampered reconstruction efforts later when quantities of water were needed for mixing with materials for house building. Likewise, earthquake damage to mountain foot trails curtailed vital access to food and markets for vulnerable households, and the widespread damage to schools threatened the long term educational prospects of many children.

The post-disaster recovery phase presents communities with a significant opportunity to 'build back better' with stronger, safer and more disaster resilient infrastructure and systems. As well as good quality and resilient construction design and implementation, this includes effective asset management, operation, monitoring and maintenance which can reduce the risk the risk of failure due to future changes in environmental and climactic conditions.

Purnima rebuilt infrastructure damaged by the earthquake while ensuring that the infrastructure would bring improved services to communities, such as water services at the household level to reducing water fetching time to zero, and school buildings constructed with adequate facilities including playgrounds and WASH facilities. Design and construction of all the infrastructures were made climate-resilient to ensure long term sustainability.

5.1 Infrastructure quality and delivery

Building local institutional capacity for quality: Quality is a key factor for sustainability, but local governments in Nepal face multiple challenges to maintain and ensure quality including lack of technical staff and expertise on design and supervision, inadequate budget allocation due to the large unmet needs of communities and hence resources are spread too thinly. Poor coordination mechanisms among various development organizations and projects, especially regarding roads causes landslides and destroys the water systems and water sources. We helped to develop capacity of the Palika technical teams to design and supervise the infrastructure projects so that quality can be maintained. We also helped them to develop infrastructure selection procedures and procurement plans so that adequate budget can be allocated for quality and timely completion.

Choosing the most effective implementation arrangement: Use of private contractors, NGOs and user committee models are the prevalent implementation modalities for small scale rural infrastructure delivery in Nepal. Purnima used all three models depending on the level of support required:

- **Contractors** with results-based payment following a competitive bidding process was the most effective modality for more technically demanding engineering projects such as trail bridges and management of foot trail construction, as they could deliver quickly and be financially held to account for quality.
- We worked with a mix of **international and local NGOs** to deliver rural water supply schemes as they had the most capacity to deliver at scale after the earthquake when re-establishing water access was urgent. It also encouraged community awareness and inclusion during the design and implementation as NGO processes had stronger inclusion requirements. However, NGO management and overhead costs can be high and there is a risk of undermining local capacity and agency in the decision-making process, particularly in project selection.

- The local government mostly employs **user committees** in the implementation of small projects (up to 6 million NPR). The users committee model for schemes such as water supply and irrigation are ideal as they are the beneficiaries and are motivated to operate and maintain the infrastructure, but a lot of support is required. Purnima provided pre-construction and post construction training to user committees for various types of infrastructure. The user committee model for community managed schemes is highly relevant for long term sustainability but requires time and investment.

5.2 Sustainability and maintenance

Sustainable management and rehabilitation: We used a sustainability framework to define required actions for all the relevant stakeholders - community, local government and implementing organisations. This prioritised handing over the long-term asset management including major rehabilitation and replacement to local stakeholders. While daily operation and maintenance of infrastructure can be managed by local user committees, higher cost repairs and rehabilitation should be financed by the local authorities. In the case of water supply, we helped Palikas develop a model water board act and directives laying out their responsibilities in terms of asset insurance, maintenance and repairs.

Climate resilience: Climate resilience was a key concern of Purnima. Purnima's WASH manual gives details of the climate resilient design parameters, training to the partners, and to the users committees. The user's committees in turn prepare climate resilience water safety plan and actions for the entire water systems for a long run. The system approach ensures community resilience for long lasting effect. The monitoring check list also contains climate resilience indicators to ensure its actual implementation on ground. Climate induced disaster as well is considered in the design and the site location of the infrastructures. In the case of WASH, source protection, recharging mechanism, bi-engineering, drainage, earthquake resistant design, avoidance of structures from the climate induced landslides, floods are important consideration as part of climate resilient. For school buildings, earthquake resistance design and construction site avoiding climate induced landslide and flood and geotechnical risk areas were important climate resilience considerations. For foot trails, the routes avoided the climate induced landslide and flood areas.

5.3 Gender equality and social inclusion

Inclusion begins at the identification stage: Purnima used vulnerability criteria as a key criterion for infrastructure selection and prioritization. Previously neglected by the local government, through a series of discussion with the GP official and representatives, the vulnerability criteria were eventually agreed and accepted by the local government. The other criteria were hardship, priority of the local government, damage by the earthquake and value for money. This has resulted in reaching vulnerable people with infrastructure services.

Local employment: We used local employment targeting women and other vulnerable people during the construction activities. A total of £1.3 m was paid to local people; 37% were women. The earthwork excavation (for pipe line trenches in WASH) which was normally supposed to be community contribution in kind in the past, was assigned to women by the community. We changed policy and made this paid work so that women are paid as part of their local employment.

Yard connection with tap stands for gender inclusion: providing a tap stand to each household reduced the water fetching time to zero. In the past, the fetching time was minimum half an hour and several hours in some cases. This work was mostly done by women, with opportunity costs for other economic activities. The yard connection has been helpful to free their time and hence allow them to engage in kitchen gardening, caring for children,

participating in community groups, and income generation activities. Most importantly, community ownership has significantly increased use of taps at the yard.

Pro-poor Tariff System: In the past each household paid a flat rate, but we introduced volumetric tariffs with a banded system so that poor households could access at least a safe minimum amount of water at low cost. Those who want to use more than the standard, particularly for productive purposes such as horticulture or livestock can do so by paying a higher tariff. This reduces wastage and maximises the value of water. Additional measures were applied by some user committees to ensure that extremely poor households could access the minimum amount required for health and hygiene and wellbeing.

